

Assignment3: 2nd Order Pivotal Bootstrap

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set.seed(123)

#function to return unbiased sample's sd
sd_sample <- function(x)
{
  n <- length(x)
  if(n <= 1)
  {
    return(0)
  }
  sd(x)
}

#bootstrap CI for mean
bootstrap_ci <- function(x , alpha = 0.05 , nboot = 10000)
{
  x <- as.numeric(x)
  n <- length(x)
  mean_x <- mean(x)
  sd_samp <- sd_sample(x)

  if(n <= 1 || sd_samp == 0)
  {
    return(c(mean_x , mean_x))
  }

  #creating indices matrix
  index <- matrix(sample(1:n , size = n*nboot , replace=TRUE) , nrow=n , ncol=nboot)

  #every column of the matrix is a bootstrap sample
  x_cols <- matrix(x[index], nrow=n, ncol=nboot)

  #mean and sd of bootstrap samples
  mean_boot <- colMeans(x_cols)
  sd_boot <- apply(x_cols , 2 , sd_sample)

  #every bootstrap's standard error
  se_boot <- sd_boot/sqrt(n)
  se_boot[se_boot == 0] <- Inf

  #t-test
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t_test <- (mean_boot - mean_x) / se_boot

# quantiles for pivotal interval
q_low <- quantile(t_test , probs = alpha/2 , na.rm = TRUE , type = 7)
q_high <- quantile(t_test , probs = 1 - alpha/2 , na.rm = TRUE , type = 7)

#CI
se_x <- sd_samp/sqrt(n)
c(mean_x-q_high*se_x , mean_x-q_low*se_x)
}

#Normal/Central_Limit_Theorem CI for mean
normal_ci_mean <- function(x , alpha = 0.05)
{
  x <- as.numeric(x)
  n <- length(x)
  x_mean <- mean(x)
  sd_samp <- sd_sample(x)
  if (n <= 1 || sd_samp == 0)
  {
    return(c(x_mean, x_mean))
  }
  z <- qnorm(1 - alpha / 2)
  half <- z * sd_samp / sqrt(n)
  c(x_mean - half, x_mean + half)
}

coverage_indicators <- function(n , alpha , mu=0 , sigma=1 , nboot=10000)
{
  x <- rlnorm(n , meanlog=mu , sdlog=sigma)
  true_mean <- exp(mu + 0.5 * sigma^2)

  ci_norm <- normal_ci_mean(x , alpha=alpha)
  ci_boot <- bootstrap_ci(x , alpha=alpha , nboot=nboot)

  cov_norm <- as.numeric(ci_norm[1] <= true_mean && true_mean <= ci_norm[2])
  cov_boot <- as.numeric(ci_boot[1] <= true_mean && true_mean <= ci_boot[2])
  c(cov_norm=cov_norm , cov_boot=cov_boot)
}

#running bootstrap
run_bootstrap <- function(n , alpha , nsim=1000 , nboot=10000 , mu=0 , sigma=1)
{
  cov_norm_count <- 0
  cov_boot_count <- 0

  for (i in 1:nsim)
  {
    res <- coverage_indicators(n=n , alpha=alpha , mu=mu , sigma=sigma , nboot=nboot)

```

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    cov_norm_count <- cov_norm_count + res["cov_norm"]
    cov_boot_count <- cov_boot_count + res["cov_boot"]
  }

  cov_norm <- cov_norm_count / nsim
  cov_boot <- cov_boot_count / nsim

  se_norm <- sqrt(cov_norm * (1 - cov_norm) / nsim)
  se_boot <- sqrt(cov_boot * (1 - cov_boot) / nsim)

  data.frame(
    #dist = paste0("Lognormal(" , mu , "," , sigma , ")"),
    n = n,
    alpha = alpha,
    nominal_coverage = 1 - alpha,
    coverage_normal = cov_norm,
    mc_se_normal = se_norm,
    coverage_boot_t = cov_boot,
    mc_se_boot_t = se_boot,
    #nsim = nsim,
    #nboot = nboot,
    stringsAsFactors = FALSE
  )
}

nsim <- 1000
nboot <- 10000
mu <- 0
sigma <- 1

r1 <- run_bootstrap(n=3 , alpha=0.10 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)
r2 <- run_bootstrap(n=10 , alpha=0.10 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)
r3 <- run_bootstrap(n=30 , alpha=0.10 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)
r4 <- run_bootstrap(n=100 , alpha=0.10 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)

r5 <- run_bootstrap(n=3 , alpha=0.05 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)
r6 <- run_bootstrap(n=10 , alpha=0.05 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)
r7 <- run_bootstrap(n=30 , alpha=0.05 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)
r8 <- run_bootstrap(n=100 , alpha=0.05 , nsim=nsim , nboot=nboot , mu=mu , sigma=sigma)

result <- rbind(r1 , r2 , r3 , r4 , r5 , r6 , r7 , r8)

result$coverage_normal <- round(result$coverage_normal , 3)
result$coverage_boot_t <- round(result$coverage_boot_t , 3)
result$mc_se_normal <- round(result$mc_se_normal , 3)
result$mc_se_boot_t <- round(result$mc_se_boot_t , 3)
result$nominal_coverage <- round(result$nominal_coverage , 3)

print(result , row.names = FALSE)

##      n alpha nominal_coverage coverage_normal mc_se_normal coverage_boot_t

```

##	3	0.10	0.90	0.641	0.015	0.732
##	10	0.10	0.90	0.762	0.013	0.859
##	30	0.10	0.90	0.844	0.011	0.884
##	100	0.10	0.90	0.873	0.011	0.886
##	3	0.05	0.95	0.679	0.015	0.758
##	10	0.05	0.95	0.816	0.012	0.913
##	30	0.05	0.95	0.879	0.010	0.917
##	100	0.05	0.95	0.902	0.009	0.932
##	mc_se_boot_t					
##		0.014				
##		0.011				
##		0.010				
##		0.010				
##		0.014				
##		0.009				
##		0.009				
##		0.008				